

Pot experiment physiology update

Trillium Trail group meeting

06.19.2026

Model fitting notes

- LMEMs include:
 - Individual fixed effect terms for PlantGMTrt, ExpSoilSource, and ExpFungSource
 - All possible interaction terms
 - Wet rhizome weight as additional fixed effect to account for size
 - Better way to do this? Dry aboveground biomass?
 - Random intercept terms for field plot and photosynthesis machine

Net photosynthesis (A_{net})

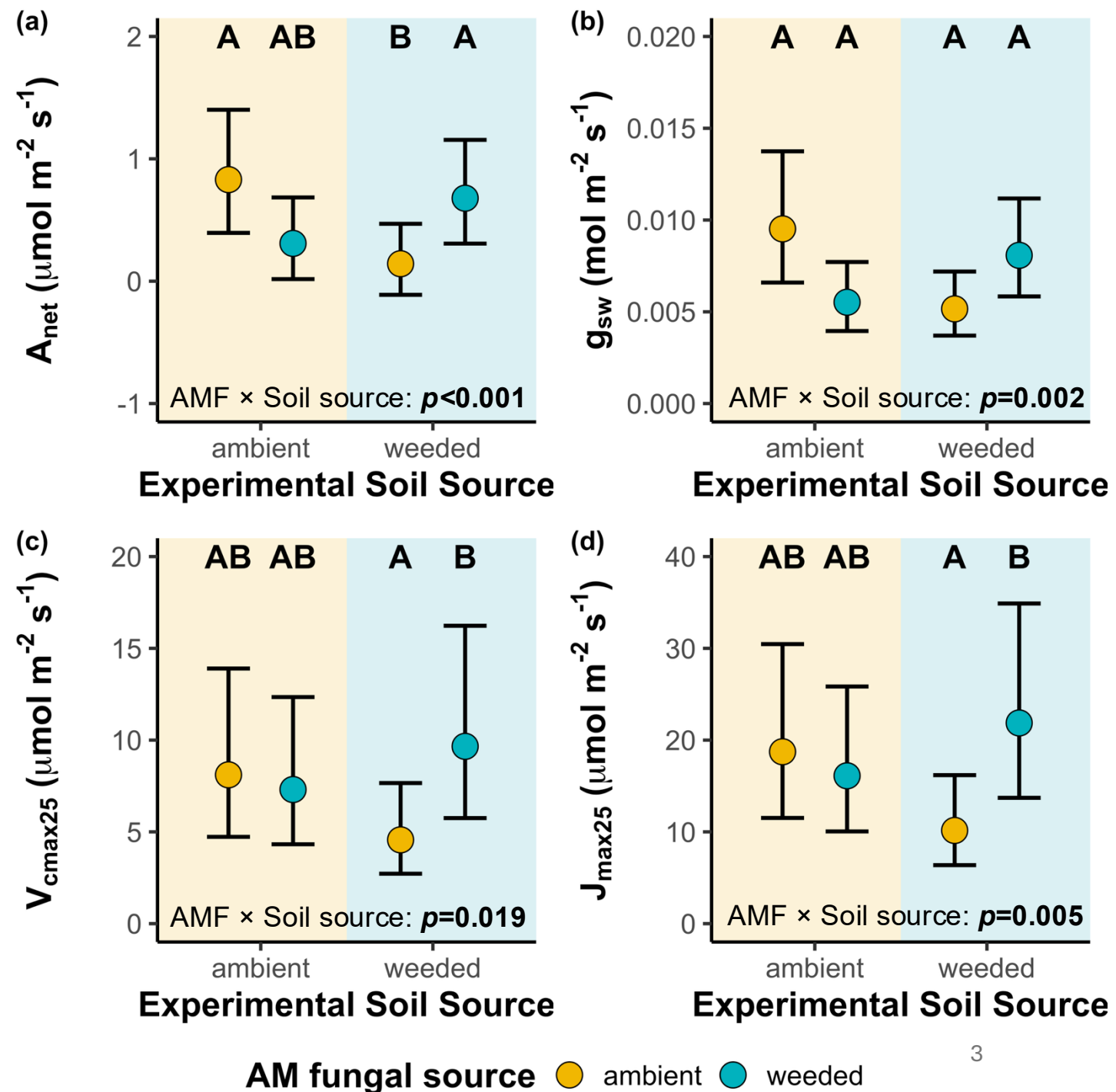
- No individual treatment effect
- Weeded AMF increases net photosynthesis, but only when plants were grown in weeded soils
- Weeded AMF also only increased net photosynthesis in plants with legacy of growing in weeded plots

Stomatal conductance (g_{sw})

- No individual treatment effect
- Stomatal conductance is lower in plants grown in weeded soils, but only for those inoculated with ambient AMF

Photosynthetic capacity (V_{cmax25} , J_{max25})

- Weeded AMF increased V_{cmax25} and J_{max25} , but only when plants were grown in weeded soils
- Weeded AMF also increased parameters when plants had legacy of growing in weeded plots



Leaf N content (N_{area})

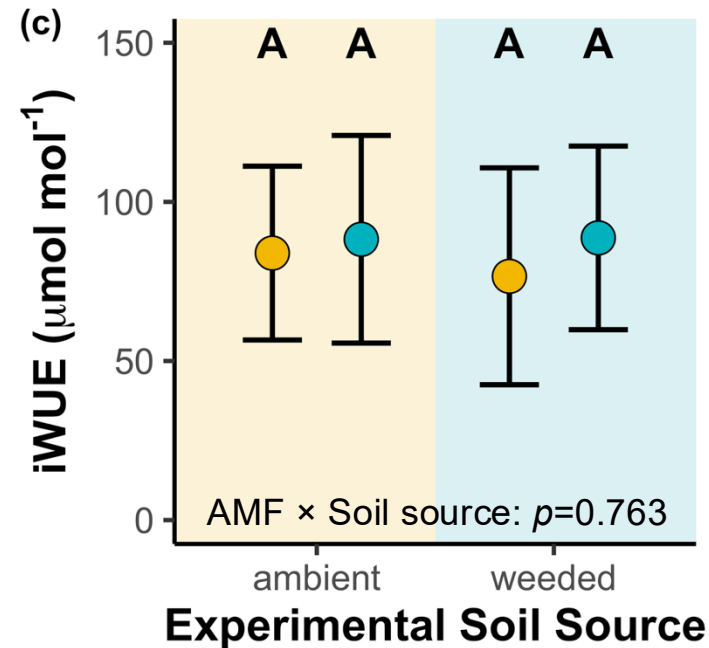
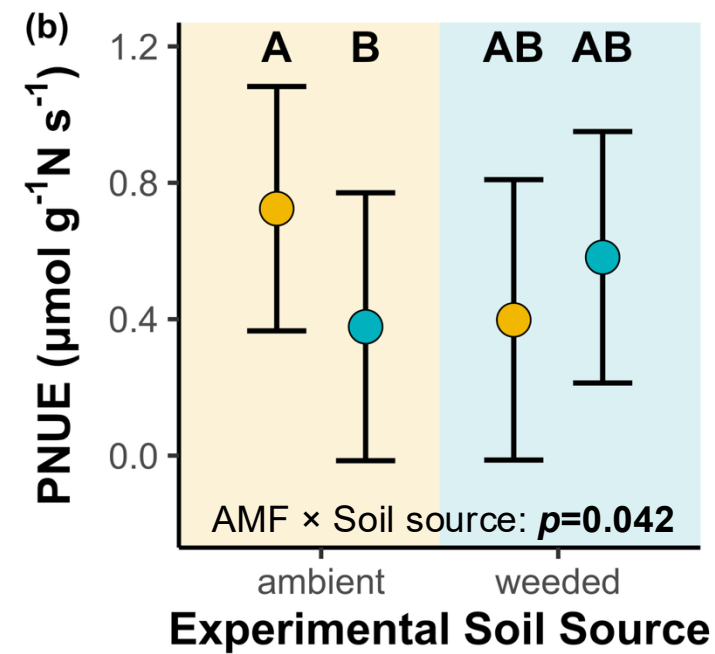
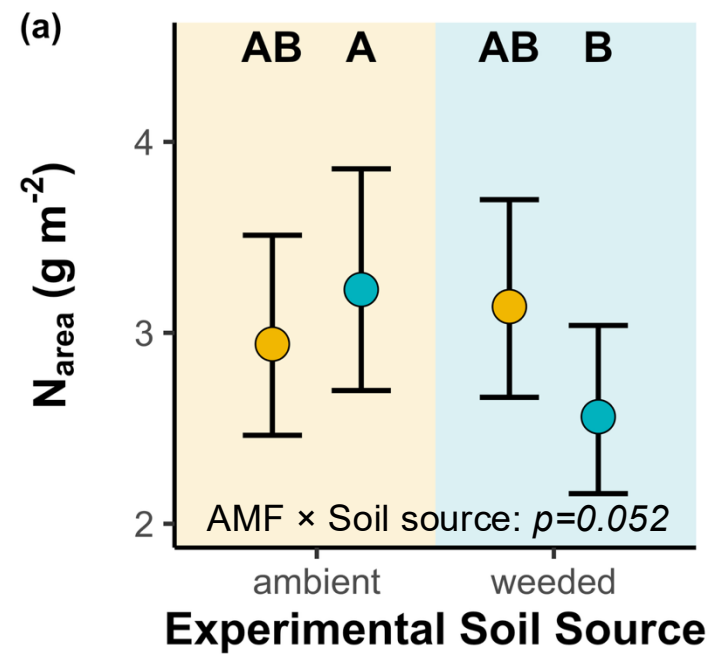
- No individual treatment effect
- Leaf N content was greater when plants were grown in ambient soils, but only when inoculated with weeded AMF or when plants had ambient treatment legacy

Photosynthetic nitrogen-use efficiency (PNUE)

- No individual treatment effect
- Weeded AMF increased PNUE in weeded soils; ambient AMF increased PNUE in ambient soils
 - Three-way interaction: ExpSoilSource * ExpFungSource interaction only observed in plants with weeded treatment legacy

Intrinsic water-use efficiency (iWUE)

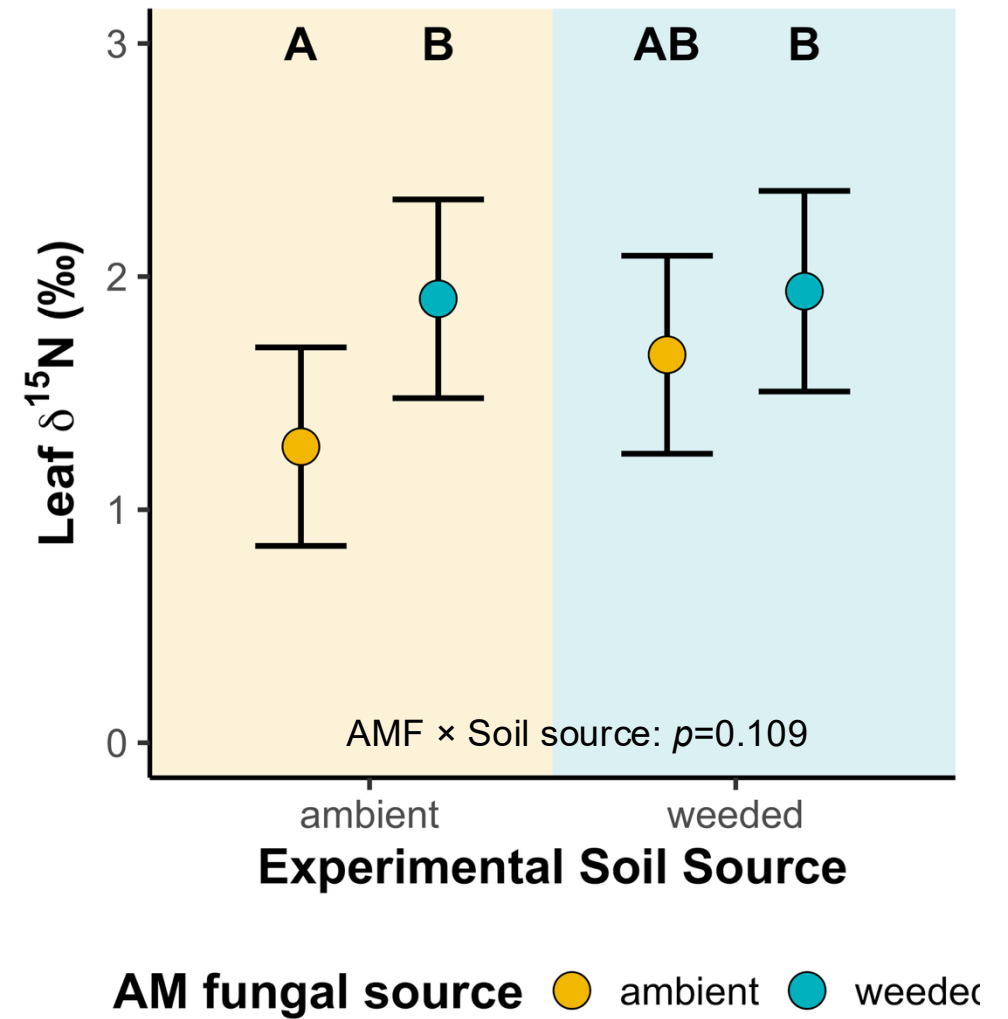
- No individual or interaction effect



AM fungal source ● ambient ● weeded

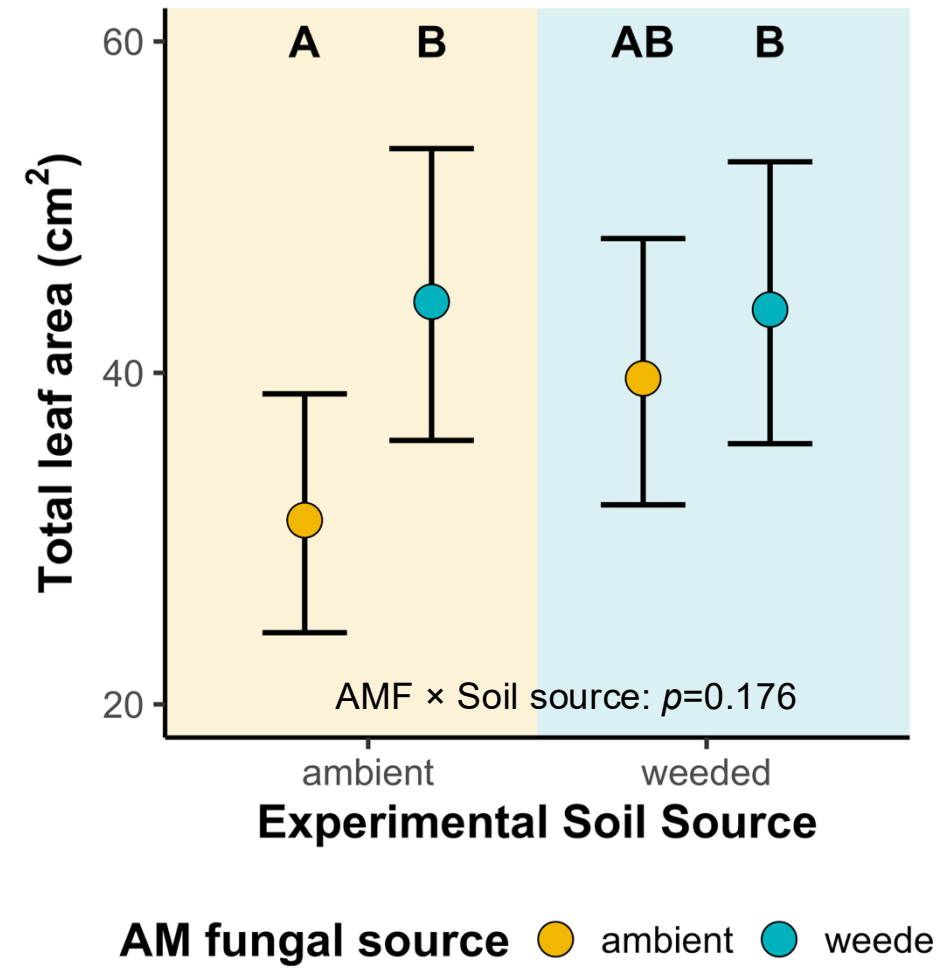
Leaf $\delta^{15}\text{N}$

- Weeded AMF increased leaf $\delta^{15}\text{N}$
- Plants grown in weeded soils have marginally greater leaf $\delta^{15}\text{N}$ ($p=0.061$)



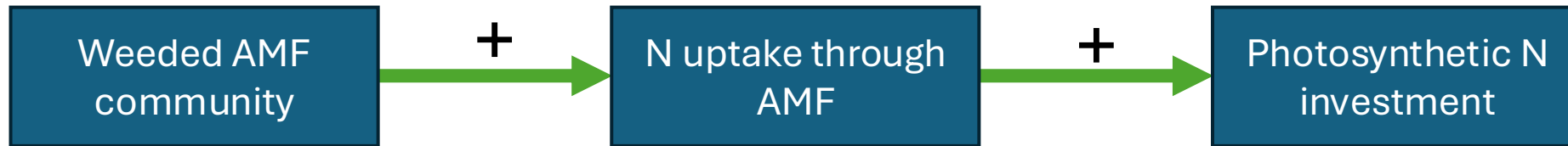
Total leaf area

- Weeded AMF increase total leaf area
- Marginal three-way interaction: weeded AMF increased total leaf area in ambient soils only, but this interaction was also only observed in plants with weeded treatment legacy



Some takeaways

- Photosynthetic and leaf $\delta^{15}\text{N}$ responses are coordinated



- Photosynthetic responses are predominately nutrient-driven
 - Strong leaf N, photosynthetic capacity treatment responses, limited stomatal conductance and water-use efficiency treatment responses
 - Same as field observations
- Soil source plays an important role in shaping fungal source responses. Soil characteristics will help clarify mechanism